

Requirement Gathering

Credit Card Approval Prediction System | Requirement Analysis Phase

Purpose

This document provides a clear understanding of the project objectives and forms the foundation for designing, developing, testing, and demonstrating the Credit Card Approval Prediction System.

Functional Requirements

ID	Requirement
FR-1	The system shall accept 17 applicant features via a web form: CODE_GENDER, FLAG_OWN_CAR, FLAG_OWN_REAL_ESTATE, FLAG_OWN_BUSINESS, FLAG_OWN_OTHER, FLAG_OWN_OTHER2, FLAG_OWN_OTHER3, FLAG_OWN_OTHER4, FLAG_OWN_OTHER5, FLAG_OWN_OTHER6, FLAG_OWN_OTHER7, FLAG_OWN_OTHER8, FLAG_OWN_OTHER9, FLAG_OWN_OTHER10, FLAG_OWN_OTHER11, FLAG_OWN_OTHER12, FLAG_OWN_OTHER13, FLAG_OWN_OTHER14, FLAG_OWN_OTHER15, FLAG_OWN_OTHER16, FLAG_OWN_OTHER17.
FR-2	The system shall load a pre-trained full scikit-learn Pipeline (best_credit_card_model.pkl) that includes preprocessing (StandardScaler, OneHotEncoder, LogisticRegression) and a final prediction step.
FR-3	The system shall predict binary outcome: 0 = Approved, 1 = Rejected.
FR-4	The system shall display the result on a dedicated result page (result.html) with a green tick (Approved) or red cross (Rejected).
FR-5	The system shall support a user-friendly age/experience input mode (years) and convert to DAYS_BIRTH / DAYS_EMPLOYED.
FR-6	The system shall support a raw days input mode (toggle) for technical users.
FR-7	The system shall fill missing OCCUPATION_TYPE with 'Unknown' to match the training pipeline's OneHotEncoder (handle_unknown='ignore').
FR-8	The system shall handle all exceptions and display an error card on result.html.

Non-Functional Requirements

ID	Requirement
NFR-1	Usability — two input modes (friendly years / raw days) for different users.
NFR-2	Performance — prediction returns in <2 s; full Pipeline handles preprocessing.
NFR-3	Portability — runs locally via Flask; deployed on Render via Gunicorn.
NFR-4	Maintainability — full Pipeline.pkl means app.py needs no preprocessing code.
NFR-5	Scalability — train_model.py allows retraining with updated data at any time.

Data Requirements

Dataset	Records	Columns	Role
application_record.csv	438,557	18	Applicant demographics and financial profile; 134,203 missing values
credit_record.csv	1,048,575	3	Monthly credit STATUS per applicant ID; used to derive the target variable

Target Variable Derivation

STATUS codes 1, 2, 3, 4, 5 in credit_record.csv indicate late/defaulted payments. Any applicant with at least one such STATUS in their history is labelled TARGET=1 (Reject). All others are TARGET=0 (Approve).